

Short-term effects of atmospheric particulate matter on myocardial infarction: a cumulative meta-analysis

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Received: 18 August 2015 / Accepted: 27 January 2016 / Published online: 5 February 2016
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Abstract Atmospheric particulate matter (PM) is hypothesized to increase the risk of myocardial infarction (MI). However, the epidemiological evidence is inconsistent. We identified 33 studies with more than 4 million MI patients and applied meta-analysis and meta-regression to assess the available evidence. Twenty-five studies presented the effects of the PM level on hospitalization for MI patients, while eight studies showed the effects on mortality. An increase in PM₁₀ was associated with hospitalization and mortality in myocardial infarction patients (RR per 10 $\mu\text{g}/\text{m}^3 = 1.011$, 95 % CI 1.006–1.016; RR per 10 $\mu\text{g}/\text{m}^3 = 1.008$, 95 % CI 1.004–1.012, respectively); PM_{2.5} also increased the risk of

hospitalization and mortality in MI patients (RR per 10 $\mu\text{g}/\text{m}^3 = 1.024$, 95 % CI 1.007–1.041 for hospitalization and RR per 10 $\mu\text{g}/\text{m}^3 = 1.012$, 95 % CI 1.010–1.015 for mortality). The results of the cumulative meta-analysis indicated that PM₁₀ and PM_{2.5} were associated with myocardial infarction with the addition of new studies each year. In conclusion, short-term exposure to high PM₁₀ and PM_{2.5} levels revealed to increase risk of hospitalization and mortality for myocardial infarction. Policy support of pollution control and individual protection was strongly recommended.

Keywords Particulate matter · Myocardial infarction · Hospitalization · Mortality · Cumulative meta-analysis

Responsible editor: Philippe Garrigues

Electronic supplementary material The online version of this article (doi:10.1007/s11356-016-6186-3) contains supplementary material, which is available to authorized users.

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Introduction

Air pollution exposure is an established risk factor for cardiovascular morbidity and mortality (Brook et al. 2010; Langrish et al. 2012). Atmospheric particulate matter (PM) is a common type of air pollution, with well-documented adverse health effects ranging from respiratory symptoms to an increased risk of hospital admissions (Brook et al. 2010; Stieb et al. 2009). Many epidemiology researchers (Bard et al. 2014; Barnett et al. 2006; Bateson and Schwartz 2004; Berglund et al. 2009, 2010; Bhaskaran et al. 2011; Braga et al. 2001; Cendon et al. 2006; Cheng et al. 2009; Dai et al. 2014; Gardner et al. 2014; Hsieh et al. 2010; Lanki et al. 2006; Linn et al. 2000; Mann et al. 2002; Mate et al. 2010; Milojevic et al. 2014; Nuvolone et al. 2011; Peters et al. 2001; Pope et al. 2006; Rich et al. 2010; Sharovsky et al. 2004; Stieb et al. 2009; Sullivan et al. 2005; Talbott et al. 2014; Ueda et al. 2009; von Klot et al. 2005; Wichmann et al. 2014; Xie et al. 2014; Zanobetti et al. 2009; Zanobetti and Schwartz 2005, 2006, 2009) have shown associations

between airborne PM and short-term risks of morbidity and mortality in patients with acute myocardial infarction (MI), but the results remain inconsistent.

In 2009, Bhaskaran et al. (Bhaskaran et al. 2009) published a systematic review, and they reported that three fifths of studies reported statistically significant detrimental effects of PM_{2.5} (PM with an aerodynamic diameter no more than 2.5 µm) on the incidence of MI, while three tenths of studies reported similar effects of PM₁₀ (PM with an aerodynamic diameter no more than 10 µm). In 2012, Mustafic et al. (Mustafic et al. 2012) reported an excellent meta-analysis in JAMA. The authors analysed the association between main air pollutants and myocardial infarction, and carbon monoxide, nitrogen dioxide, sulphur dioxide and particle matter were also analysed in this study. Their results indicated that PM_{2.5} and PM₁₀ were significantly associated with MI. However, several high-quality studies have been published from 2012 to 2015, and the Mustafic et al. study also had some limitations, such as no separate discussion of hospitalization and mortality and no adjustment for the bias introduced by including studies with different designs. However, particle matter damage to health has attracted more attention in recent years.

Therefore, we thought that it was meaningful to update and deepen the meta-analysis of the association between particle matter and myocardial infarction. We used a meta-analytic approach to combine information across individual study results to estimate an overall effect of atmospheric particulate matter on the risk of hospitalization and mortality in myocardial infarction patients.

Method

Search strategy

We identified papers through electronic searches of medical and biological databases (Medline, 1966–January 2015; Embase, 1980–January 2015; CAB Health, 1973–January 2015; the Science Citation Index and Web of Science, 1981–January 2015; and the Conference Papers Index through the Cambridge Scientific Abstracts databases, 1982–January 2015) using a comprehensive list of keywords and phrases: air pollution/particulate matter/PM_{2.5}/PM₁₀ and cardiovascular disease/myocardial infarction. We also searched web-based resources (available with the electronic versions of these articles at www.healtheffects.org), scanned reference lists of the review articles and published studies and consulted with key experts in the field. The identified publications were scanned by two of the authors (Xianlei Cai and Mengyao Tang). A flowchart of the search strategy is shown in Fig. 1.

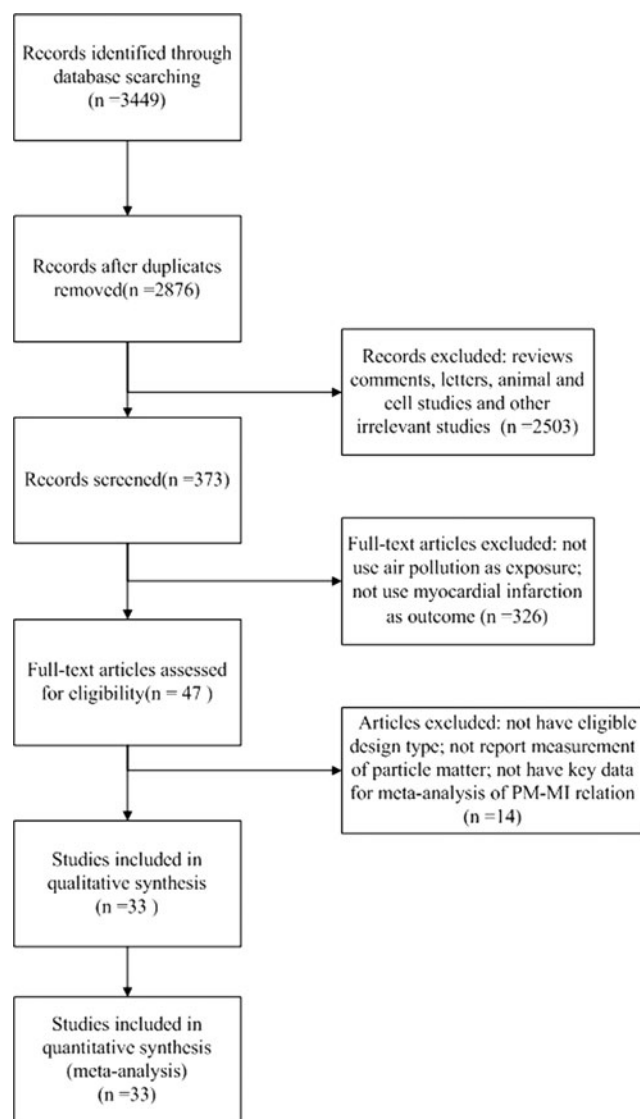


Fig. 1 A flowchart of the search strategy

Inclusion criteria

The eligible studies were selected by two investigators (Xianlei Cai and Mengyao Tang) independently, according to the following inclusion criteria. (1) The studies provided numerical estimates of the relationship between atmospheric particulate matter and myocardial infarction, as well as an indication of the uncertainty of the central estimate (e.g. 95 % confidence interval), which included the odds ratio (OR), risk ratio (RR), hazard ratio (HR) or the increased percentage of myocardial infarction morbidity and mortality (β). (2) The articles provided short-term effects and applied the case-crossover or time-series design. The events that occurred after the lag time of PM exposure ranged from 0 to 7 days which was confirmed as the short-term effects of PM. (3) The atmospheric particulate matter (PM₁₀, PM_{2.5}) concentrations were calculated through successive 24-h monitoring at the

respective monitoring stations. (4) The discharge diagnosis was classified according to the International Classification of Disease, 9th Revision (ICD-9, World Health Organization 1975, which was used until July 1999) or the 10th Revision (ICD-10, World Health Organization 1992), with the following myocardial infarction codes: ICD-9 (410) and ICD-10 (I21, I22).

Date extraction

Using a standardized form, the characteristics of the relevant articles were independently extracted by two reviewers (Xianlei Cai and Mengyao Tang), as follows: first author's name, publication year, country, study design, number of patients, type of exposure, outcome (hospitalization or death), estimates, lag time and controlled factor. When there was disagreement, a third reviewer (Xiuyang Li) made the final decision. If the study provided more than one result for the different lags, the one with the shortest lag was chosen. When necessary, the effect estimates were recalculated into RR, with a per 10 $\mu\text{g}/\text{m}^3$ increase in particulate matter levels, as described in our previous studies (Li et al. 2012; Yu et al. 2014).

Statistical analyses

Pooled RRs with 95 % confidence intervals (CI) were calculated using the fixed or random effects meta-analysis, with Q test used as the chosen measure of heterogeneity. When the Q test results showed evidence of heterogeneity ($P \leq 0.05$), we used a random effects model, following the method of DerSimonian and Laird (1986). Otherwise, a fixed-effects analysis was conducted using the method of Mantel and Haenszel (1959). Meta-regression was used to examine the sources of heterogeneity, and we considered the study design (case-crossover or time-series), location (America, Europe or Asia) and lag time. We also produced forest plots to show RRs from each of the individual studies contained in the meta-analyses and the estimation of the pooled RR. The sizes of the markers of each RR in the plots represent the relative weight each study contributed to the pooled estimation. Publication bias was assessed with a weighted Egger's test (Egger et al. 1997) and funnel plots. If there were fewer than ten studies in the pooled analysis, we did not conduct the meta-regression. The Egger's test and funnel plots were also not considered when the results of the pooled analysis were based on a limited number of small trials. In addition, we performed a cumulative meta-analysis to summarize the evidence in the assessment year of the selected studies. In order publication year, the studies were added one at a time to calculate the results sequentially. All analyses were performed using software STATA version 12.0 (StataCorp LP, College Station, TX, USA).

Results

Summary results

The systematic search identified 33 studies (19 case-crossover studies and 14 time-series studies) that fulfilled the inclusion criteria. Twenty-five studies demonstrated the effects of PM on hospitalization for MI, while eight studies showed the effects of PM on mortality. Eighteen studies investigated the relationship between PM_{10} and MI, 12 studies researched the relationship between $\text{PM}_{2.5}$ and MI, and 3 studies investigated the effects of PM_{10} and $\text{PM}_{2.5}$ concurrently. These studies reported 41 independent estimates, with more than 4,000,000 MI patients. The lag time varied from 0 to 7 days. Table 1 shows the summaries and results from each study included in this meta-analysis. Supplementary Table 1 provides other data, such as the average PM, mean age and percentage of men.

The association between PM_{10} and MI

In total, 17 estimates of the relationship between PM and MI hospitalization were incorporated into the meta-analysis. We included the study location, design and lag time in the meta-regression analysis to examine any possible heterogeneity (location, $P=0.217$; design, $P=0.720$; lag time, $P=0.624$), and we found that the study location, design and lag time did not explain any of the observed heterogeneity. The pooled RR was 1.011 (95 % CI 1.006–1.016) with heterogeneity ($Q=48.22$; $P=0.000$); a forest plot and a funnel plot are shown in Fig. 2 and Fig. S1(A), respectively. The Egger's test did not reveal any evidence of publication bias ($t=1.88$; $P=0.079$). Figure S2 shows the result of the cumulative meta-analysis; we found that since 2005, the results began to show significance, with a final result of 1.011 per 10 $\mu\text{g}/\text{m}^3$ increase in PM_{10} level.

Four estimates of the relationship between PM and MI mortality were included in the meta-analysis. The pooled RR was 1.008 (95 % CI 1.004–1.012), with homogeneity ($Q=6.22$; $P=0.102$). A forest plot is shown in Fig. 3. The results of the cumulative meta-analysis indicated that PM_{10} was always associated with MI mortality (as shown in Fig. S3).

The association between $\text{PM}_{2.5}$ and MI

We conducted a meta-analysis for 16 estimates of the relationship between PM and MI hospitalization. The meta-regression analysis did not detect any heterogeneity that was explained by location, design and lag time (location, $P=0.183$; design, $P=0.309$; lag time, $P=0.257$); therefore, we incorporated these 16 estimates into the meta-analysis. The pooled RR was 1.024 (95 % CI 1.007–1.041), with heterogeneity ($Q=97.96$, $P=0.000$); a forest plot and a funnel plot are

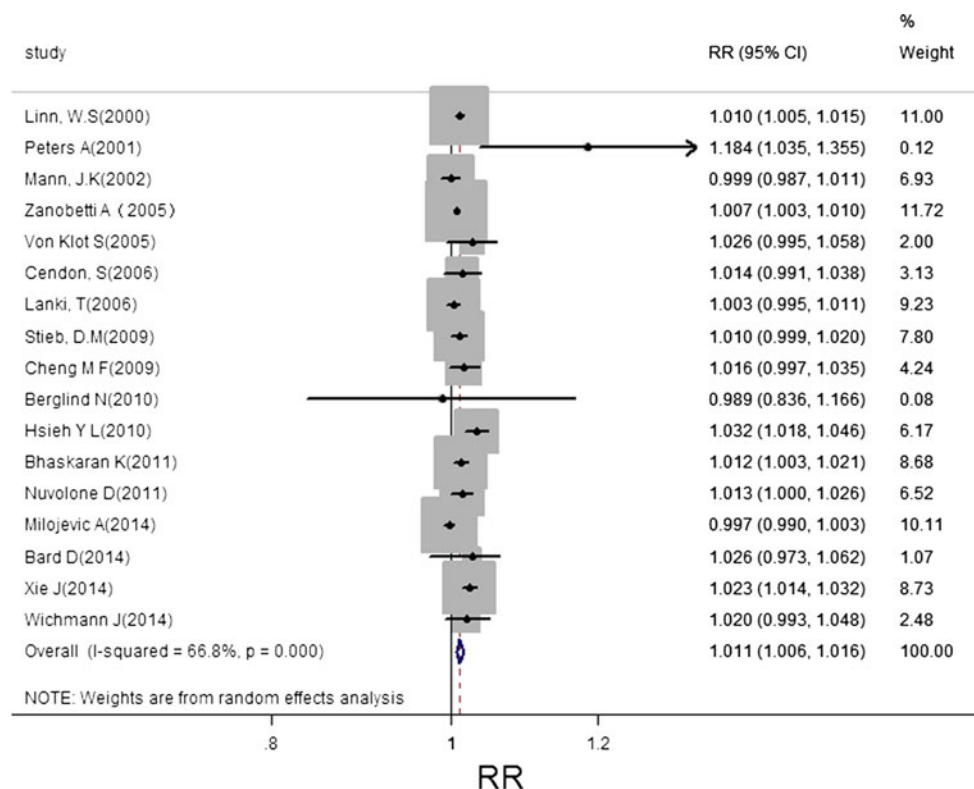
Table 1 Characteristics of studies included for meta-analysis

Study	Country	Design	Number of patients	Type of PM	Outcome	OR (95 % CI)	Lag	Controlled factor
Wichmann et al. (2014)	Sweden	Case-crossover	10,972	PM ₁₀	Hospitalization	1.020 (0.993, 1.048)	0 days	Temperature, relative humidity and public holidays
Xie et al. (2014)	China	Time-series	6200	PM ₁₀	Hospitalization	1.023 (1.014, 1.032)	0 days	Temperature and relative humidity
Bard et al. (2014)	France	Case-crossover	2134	PM ₁₀	Hospitalization	1.026 (0.973, 1.062)	0 days	Holidays, temperature, atmospheric pressure and mean relative humidity
Milojovic et al. (2014)	England and Wales	Case-crossover	365,151	PM ₁₀	Hospitalization	0.997 (0.990, 1.003)	0–4 days	Ambient temperature and day of week
Nuvolone et al. (2011)	Italy	Case-crossover	11,450	PM ₁₀	Hospitalization	1.013 (1.000, 1.026)	2 days	Age, gender, body mass index, time-dependent variables
Bhaskaran et al. (2011)	England	Case-crossover	79,288	PM ₁₀	Hospitalization	1.012 (1.003, 1.021)	1–6 h	Ambient temperature, relative humidity, circulating levels of influenza and respiratory syncytial virus and residual seasonality within calendar month strata
Hsieh et al. (2010)	Taiwan	Case-crossover	23,420	PM ₁₀	Hospitalization	1.032 (1.018, 1.046)	2 days	Temperature and humidity
Berglund et al. (2010)	Sweden	Case-crossover	660	PM ₁₀	Hospitalization	0.989 (0.836, 1.166)	1 day	Temperature and relative humidity
Cheng et al. (2009)	Taiwan	Case-crossover	9349	PM ₁₀	Hospitalization	1.016 (0.997, 1.035)	2 days	Temperature and humidity
Stieb et al. (2009)	Canada	Time-series	63,184	PM ₁₀	Hospitalization	1.010 (0.999, 1.020)	1 day	Seasonal cycles in air pollution and health outcomes
Lanki et al. (2006)	Europe	Time-series	26,854	PM ₁₀	Hospitalization	1.003 (0.995, 1.011)	0 days	N/A
Cendon et al. (2006)	Brazil	Time-series	19,272	PM ₁₀	Hospitalization	1.014 (0.991, 1.038)	0–7 days	Season and trend, humidity, day of week
von Klot et al. (2005)	Europe	Case-crossover	22,006	PM ₁₀	Hospitalization	1.026 (0.995, 1.058)	1 day	Trend and season
Zanobetti and Schwartz (2005)	USA	Case-crossover	302,453	PM ₁₀	Hospitalization	1.007 (1.003, 1.010)	0–2 days	Day of week, temperature
Mann et al. (2002)	USA	Time-series	19,690	PM ₁₀	Hospitalization	0.999 (0.987, 1.011)	0–2 days	Day of week, year and region-specific cubic B-spline smooths of study day, temperature and relative humidity
Peters et al. (2001)	USA	Case-crossover	764	PM ₁₀	Hospitalization	1.184 (1.035, 1.355)	1 day	Season, day of the week and meteorological parameters
Linn et al. (2000)	USA	Time-series	N/A	PM ₁₀	Hospitalization	1.010 (1.005, 1.015)	N/A	Season and station
Berglund et al. (2009)	Europe	Time-series	25,006	PM ₁₀	Death	1.052 (1.016–1.097)	1 day	N/A
Bateson and Schwartz (2004)	USA	Case-crossover	6157	PM ₁₀	Death	1.020 (0.998, 1.041)	0–1 day	N/A
Sharovsky et al. (2004)	Brazil	Time-series	12,007	PM ₁₀	Death	1.010 (0.910–1.110)	N/A	N/A
Braga et al. (2001)	USA	Time-series	N/A	PM ₁₀	Death	1.007 (1.003, 1.011)	2 days	N/A
Talbot et al. (2014)	Florida Massachusetts New Jersey New Hampshire	Case-crossover Case-crossover Case-crossover Case-crossover	303,405, 18,099 146,084 12,934	PM _{2.5} PM _{2.5} PM _{2.5} PM _{2.5}	Hospitalization Hospitalization Hospitalization Hospitalization	0.951 (0.933, 0.969) 1.055 (1.035, 1.075) 1.058 (1.043, 1.074) 1.034 (0.976, 1.096)	0 days 0 days 0 days 0 days	Temperature and ozone, lag and state Temperature and ozone, lag and state Temperature and ozone, lag and state Temperature and ozone, lag and state

Table 1 (continued)

Study	Country	Design	Number of patients	Type of PM	Outcome	OR (95 % CI)	Lag	Controlled factor
Milojevic et al. (2014)	New Mexico	Case-crossover	330,602	PM _{2.5}	Hospitalization	0.930 (0.846, 1.023)	0 days	Temperature and ozone, lag and state
	New York	Case-crossover	112,683	PM _{2.5}	Hospitalization	1.009 (1.003, 1.014)	0 days	
	England and Wales	Case-crossover	134,964	PM _{2.5}	Hospitalization	0.996 (0.976, 1.016)	0–4 days	Ambient temperature and day of week
Gardner et al. (2014)	USA	Case-crossover	329	PM _{2.5}	Hospitalization	1.262 (1.014, 1.574)	0 days	N/A
Rich et al. (2010)	USA	Case-crossover	5864	PM _{2.5}	Hospitalization	1.019 (0.981, 1.065)	1d	Apparent temperature during that same lag period
Stieb et al. (2009)	Canada	Time-series	400,000	PM _{2.5}	Hospitalization	1.013 (0.976, 1.054)	1 day	Seasonal cycles in air pollution and health outcomes
Zanobetti et al. (2009)	USA	Time-series	121,652	PM _{2.5}	Hospitalization	1.023 (1.011, 1.035)	2 days	Temperature
Barnett et al. (2006)	Australia New Zealand	Case-crossover	28,818	PM _{2.5}	Hospitalization	1.073 (1.035, 1.117)	1 day	Temperature, humidity, pressure, day of week, public holiday, rainfall
Pope et al. (2006)	USA	Case-crossover	1173	PM _{2.5}	Hospitalization	1.033 (0.962, 1.115)	1 day	Day of the week, season, trend, individual characteristic
Zanobetti and Schwartz (2006)	USA	Case-crossover	15,578	PM _{2.5}	Hospitalization	1.054 (1.007, 1.099)	1 day	Temperature, day of week
Sullivan et al. (2005)	USA	Case-crossover	5793	PM _{2.5}	Hospitalization	1.020 (0.980, 1.070)	1 h	Temperature
Peters et al. (2001)	USA	Case-crossover	764	PM _{2.5}	Hospitalization	1.273 (1.063, 1.530)	1 day	Season, day of the week and meteorological parameters
Dai et al. (2014)	USA	Time-series	308,235	PM _{2.5}	Death	1.012 (1.006, 1.012)	2 days	Temperature
Mate et al. (2010)	Spain	Time-series	4243	PM _{2.5}	Death	1.066 (1.032, 1.100)	6 days	N/A
Zanobetti and Schwartz (2009)	USA	Time-series	397,894	PM _{2.5}	Death	1.012 (1.005, 1.019)	2 days	N/A
Ueda et al. (2009)	Japan	Time-series	16,129	PM _{2.5}	Death	1.013 (0.992, 1.035)	0 days	Temperature, humidity, seasonality and day of the week

Fig. 2 A forest plot of the relationship between PM₁₀ and MI hospitalization



shown in Fig. 4 and Fig. S1(B), respectively. Publication bias was not detected (Egger's test, $t = 1.27$; $P = 0.225$). Figure S4 shows the result of the cumulative meta-analysis. We found that since 2006, the results began to show significance, culminating with 1.024 per 10 $\mu\text{g}/\text{m}^3$ increase in PM_{2.5} level.

Four estimates of the relationship between PM and MI mortality were included in the meta-analysis. The pooled RR was 1.012 (95 % CI 1.010–1.015), with homogeneity ($Q = 7.78$, $P = 0.051$). A forest plot is shown in Fig. 5. The results of the cumulative meta-analysis indicated that PM_{2.5}

has been associated with MI mortality since 2009, with 1.012 per 10 $\mu\text{g}/\text{m}^3$ increase in PM_{2.5} level (as shown in Fig. S5).

Discussion

Recently, many epidemiology studies have reported the relationship between particle matter and myocardial infarction, with controversies. The studies of Xie et al. (2014), (Nuvolone et al. (2011), Bhaskaran et al. (2011), Hsieh et al.

Fig. 3 A forest plot of the relationship between PM₁₀ and MI mortality

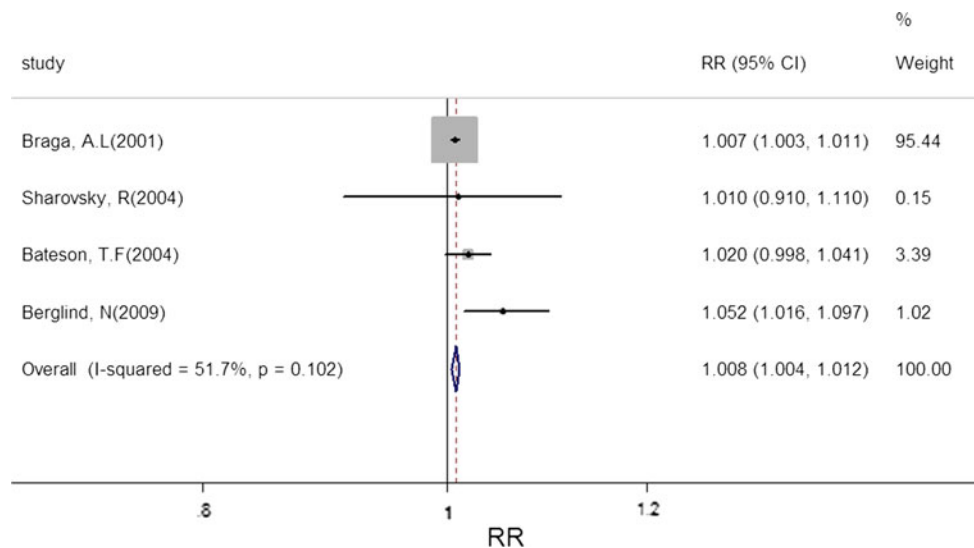
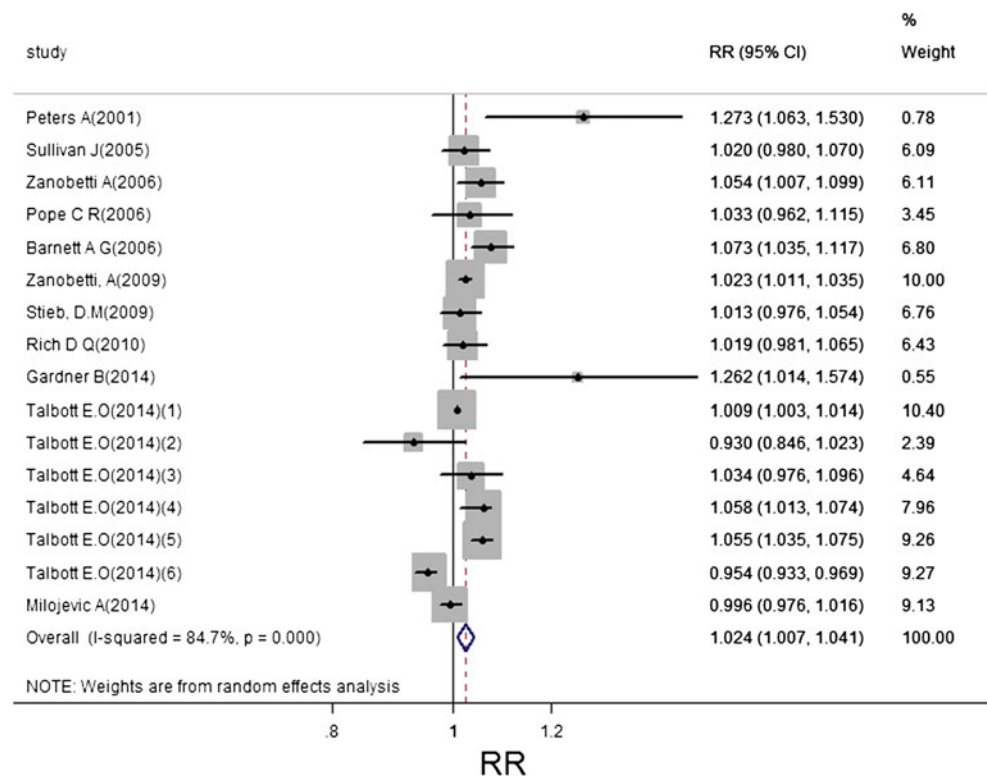


Fig. 4 A forest plot of the relationship between PM_{2.5} and MI hospitalization, and the estimates from Talbott et al. (2014) (1) belong to Florida; Talbott et al. (2014) (2) belong to Massachusetts; Talbott et al. (2014) (3) belong to New Jersey; Talbott et al. (2014) (4) belong to New Hampshire; Talbott et al. (2014) (5) belong to New Mexico; Talbott et al. (2014) (6) belong to New York

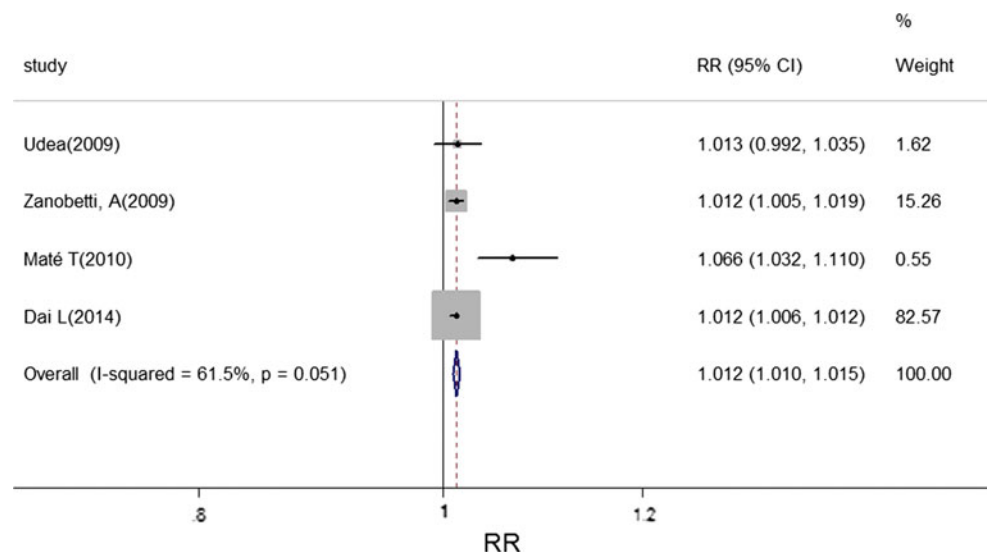


(2010), Zanobetti and Schwartz (2005), Peters et al. (2001) and Linn et al. (2000) indicated that PM₁₀ level is associated with hospitalizations for myocardial infarction, and the studies of Gardner et al. (2014), Zanobetti et al. (2009), Zanobetti and Schwartz (2006) and Barnett et al. (2006) reported that PM_{2.5} is also associated with MI morbidity. However, other studies (Bard et al. 2014; Berglind et al. 2010; Cendon et al. 2006; Cheng et al. 2009; Lanki et al. 2006; Mann et al. 2002; Milojevic et al. 2014; Pope et al. 2006; Rich et al. 2010; Stieb et al. 2009; Sullivan et al. 2005; von Klot et al. 2005) have not indicated that atmospheric particulate matter could

increase the incidence of MI. This analogous situation is found in those studies that have focused on the relationship between PM and MI mortality. Although some studies (Berglind et al. 2009; Braga et al. 2001; Dai et al. 2014; Mate et al. 2010; Zanobetti and Schwartz 2009) have identified positive associations between particulate matter and MI mortality, others have yielded null or inconclusive results (Nuvolone et al. 2011; Pope et al. 2006; Sullivan et al. 2005).

Our meta-analysis identified 33 relevant and suitable studies published from 2000 to 2014. These studies provided epidemiological evidence from more than 4 million participants

Fig. 5 A forest plot of the relationship between PM_{2.5} and MI mortality



overall. We used meta-regression to detect the influence of the study design, location and lag time, which were the three factors that likely resulted in heterogeneity, and the results suggested that the study design, location and lag time were not sources of heterogeneity. Although we found heterogeneity among the PM-MI hospitalization studies, it could be attributed to a large number of included estimates, as well as the differences in age, gender and average PM. There was no good method of adjusting for these possible confounding factors because not every study provided sufficient data of the mean age, gender or other factors; therefore, it was difficult to conduct meta-regression analyses for these factors. However, we found that most of the relevant studies included patients older than 60 years of age and researched males and females together, with similar percentages of males (Supplementary Table 1). Hence, we considered that these factors did not significantly affect the accuracy of the conclusion. In addition, we needed more data and studies to conduct further analysis and to avoid these possible sources of heterogeneity.

The evidence gained in this study showed that increased exposure to PM₁₀ and PM_{2.5} increased the risk of MI, including hospitalization (RR per 10 $\mu\text{g}/\text{m}^3 = 1.011$, 95 % CI 1.006–1.016; RR per 10 $\mu\text{g}/\text{m}^3 = 1.024$, 95 % CI 1.007–1.041, respectively) and mortality (RR per 10 $\mu\text{g}/\text{m}^3 = 1.008$, 95 % CI 1.004–1.012; RR per 10 $\mu\text{g}/\text{m}^3 = 1.012$, 95 % CI 1.010–1.015, respectively), although the summary estimates were small. The results of the cumulative meta-analysis, with the addition of new studies year by year, indicated that PM₁₀ and PM_{2.5} were associated with hospitalization and mortality in myocardial infarction patients and that the RRs tended to be stable. These results indicated that the adverse effect of atmospheric particle matter was validated constantly with the increase of new studies every year, thereby improving the accuracy and scientificity of the conclusion. Our results contribute to the understanding of PM-related health effects.

The biological mechanisms of particulate matter on myocardial infarction are not clearly elucidated at present. The following plausible mechanisms could be responsible for such particulate matter effects: telomere erosion induced by oxidative stress and inflammatory responses triggering endothelial dysfunction, atherosclerosis and thrombosis, which interferes with the autonomic nervous system (Franchini and Mannucci 2007; Grahame and Schlesinger 2012; Scapellato and Lotti 2007; Utell et al. 2002).

With the progress of industrialization and the increase in the number of private cars, a growing number of developing countries, such as China, are threatened by air pollution, particularly atmospheric particle matter. Meanwhile, these developing countries have large and aging populations. Therefore, although the pooled estimates are small, it still affects many people when considering the population base. However, myocardial infarction patients have huge economic burdens and generally poor prognoses. It is well known that percutaneous

coronary intervention (PCI) is an effective treatment for acute MI. However, stents are quite expensive for the vast majority of patients. Patients must also take medicine for prolonged periods, sometimes lifelong. Note that many patients did not have the opportunity to receive treatment for PCI because of time limitations. We think that these results are meaningful and should attract the attention of governments to restrict the emissions of particulate matter and control air pollution.

Our study has some limitations. Thirty-three studies were listed in our meta-analysis, but not every study controlled for the same or similar confounding factors. This finding could impact the results and could result in heterogeneity; however, it was hard to avoid. We regarded the lag time of PM exposure (from 0 to 7 days) as a short-term effect, and the long-term effects of particulate matter levels on myocardial infarction were not explored in our study for a limited number of pertinent articles. Because of the measurement error of different air pollution monitor apparatuses, the process of particulate matter measurement could lead to bias. We only performed four estimates for the pooled analysis of the association between PM and the mortality rate of MI. It was inappropriate to conduct meta-regression analysis and the Egger's test, and the insufficient number of relevant estimates influenced accuracy of the results. Although some factors, such as the mean age of the study population and percentage of men, might be responsible for the observed heterogeneity, we lack sufficient data to conduct meta-regression analysis of these factors additionally. We were unable to extract detailed data of average PM from 11 articles, mean age from 20 articles and percentage of men from 17 articles. Nonetheless, average PM, age and gender were not observed to be responsible for the heterogeneity for PM₁₀-MI association by meta-regression combining the studies where the information was available (Supplementary Table 1). More relevant studies and detailed data are needed to overcome these limitations.

Our study also has some strengths. First, we incorporated more studies, adjusted for different designs, locations and lag times, and analysed hospitalization and mortality rates. We also found similar but more detailed results. The cumulative meta-analysis was conducted to illustrate the variation trend of PM-MI relationships. Second, we applied strict inclusion criteria to minimize selection bias, and we conducted meta-regression analysis to identify the sources of heterogeneity between the included studies. Third, we reviewed a large number of studies, with more than 4 million participants, to improve the credibility of the conclusion. Moreover, we did not discover any evidence of publication bias using the Egger's test.

Conclusion

In the study, short-term exposure to high PM₁₀ and PM_{2.5} levels increased the risk of hospitalization and mortality in

myocardial infarction patients. Policy support of pollution control and individual protection was strongly recommended.

Contributors Liuyang Li, Zhenhong Li and E. Marian Scott conceived and designed the study; Xianlei Cai and Mengyao Tang acquired and analysed the data; Xianlei Cai, Mengyao Tang and Xiuyang Li interpreted the data; Xianlei Cai and Mengyao Tang drafted the manuscript; and Zhenhong Li and E. Marian Scott critically revised the article. All authors approved the final version to be published.

Funding This research was supported by a grant from the National Twelfth Five-year Science and Technology Support Plan Project in the Field of Population and Health (grant 2014BAI08B00) and the Zhejiang University Undergraduate Zetetic Experiment Public Health Project (2013). The funding agencies played no role in the study design, data collection and analysis, decision to publish or preparation of the manuscript.

Compliance with ethical standards

Competing financial interest The authors have declared no competing financial interests.

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